

ECM Summary from RW4 Primary Chain

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

RW4 domestic vertical chain is FarmGateUA -> ProducerUA -> ProZorro -> Retail, estimated with both forward and reverse-flow pairs. Farm-gate enters from two alternative reconstruction workbooks and both linear/pchip interpolation variants are carried end-to-end.

family=ECM

rows=7

source=/Users/getapple/Documents/KSE/Master Thesis/Main materials/Model/Charniuk_Dairy_Research/outputs/primary_chain_summary/primary_chain_consolidated.xlsx

Table: ECM_Summary

panel_level	panel_name	segment_name	product	standardized_type	retailer_panel	brand	price_variant	reconstruction_variant	farm_gate_soure	frequency	chain_direction	stage_from	stage_to	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	unreliable_reason	shock_dummy_included	promo_controls_included	ljungbox_p	arch_p	jb_p	notes	robust_linear_vs_pchip	interpolation_sensitive	robust_across_reconstruction	core_finding_flag
product	product::Сир твердый::Sil...	Сир твердый	Сир твердый	hard_cheese	Silpo	nan	observed	linear	initial	daily	reverse	Retail	FarmGateUA	retail_to_farmgateua	ECM	FarmGateUA	Retail	36	-0.0006243607074365656	-0.05118671340979519	-0.1304047954777818	0.01618224692205509	nan	nan	nan	ok	0	nan	0	0.4546591156239976	0.1381724793310779	0.03583439982485814	engle_granger_p=0.02972; ...	0	1	0		
average	average::Silpo::observed:... all_products_average	all_products_average	all_products_average	all_products_average	Silpo	nan	observed	linear	initial	daily	forward	FarmGateUA	Retail	farmgateua_to_retail	ECM	Retail	FarmGateUA	45	-0.9322772851319434	-1.56271707030248	-0.5172596333297379	3.278473445321985e-06	nan	nan	nan	ok	0	nan	0	0.8972381078358652	0.7690540042589011	0.5837497917198777	engle_granger_p=0.002921; ...	1	0	1		
product	product::Молоко питне::Si...	Молоко питне	Молоко питне	milk	Silpo	nan	baseline	linear	all_missing_filled	daily	forward	FarmGateUA	Retail	farmgateua_to_retail	ECM	Retail	FarmGateUA	38	-0.7299336643576522	-1.463438024093577	-0.7973434884556778	9.528715655827255e-06	nan	nan	nan	ok	0	nan	0	0.1144573872682583	0.3064661705975509	0.4681731907578257	engle_granger_p=0.00092; ...	1	0	0		
average	average::Silpo::observed:... all_products_average	all_products_average	all_products_average	all_products_average	Silpo	nan	observed	linear	all_missing_filled	daily	forward	FarmGateUA	Retail	farmgateua_to_retail	ECM	Retail	FarmGateUA	45	0.4518884015846381	-1.469903753919038	-0.6977591555383247	5.092795057104528e-07	nan	nan	nan	ok	0	nan	0	0.7970888264642652	0.7870655941841516	0.6313790892338038	engle_granger_p=0.000208; ...	1	0	1		
average	average::Silpo::observed:... all_products_average	all_products_average	all_products_average	all_products_average	Silpo	nan	observed	pchip	initial	daily	forward	FarmGateUA	Retail	farmgateua_to_retail	ECM	Retail	FarmGateUA	45	-0.7725216333196893	-1.449885237251225	-0.516964061678618	4.607319487298131e-06	nan	nan	nan	ok	0	nan	0	0.8977475095145333	0.8019656365888165	0.5416586765569108	engle_granger_p=0.002787; ...	1	0	1		
product	product::Молоко питне::Si...	Молоко питне	Молоко питне	milk	Silpo	nan	baseline	pchip	all_missing_filled	daily	forward	FarmGateUA	Retail	farmgateua_to_retail	ECM	Retail	FarmGateUA	39	-0.3184687744149954	-1.376078842060526	-0.7710056136386508	9.153908723126643e-05	nan	nan	nan	ok	0	nan	0	0.2986309004515825	0.7193751009761611	0.3295111856425572	engle_granger_p=9.881e-05; ...	1	0	0		
average	average::Silpo::observed:... all_products_average	all_products_average	all_products_average	all_products_average	Silpo	nan	observed	pchip	all_missing_filled	daily	forward	FarmGateUA	Retail	farmgateua_to_retail	ECM	Retail	FarmGateUA	45	0.4031624181582367	-1.450887424487711	-0.7128005383669054	2.106198918845673e-07	nan	nan	nan	ok	0	nan	0	0.801596868197212	0.7694835593364274	0.598681891822242	engle_granger_p=0.0001423; ...	1	0	1		